

Knot yet: minimum marriage age law, marriage delay, and earnings

Chunbei Wang¹ · Le Wang¹

Received: 18 March 2015 / Accepted: 6 January 2017 / Published online: 26 January 2017
© Springer-Verlag Berlin Heidelberg 2017

Abstract Despite the historical highs for age at first marriage, little is known about the causal relationship between marriage delay and wages, and more importantly, the mechanisms driving such relationship. We attempt to fill the void. Building on an identification strategy proposed in Dahl (Demography 47:689–718, 2010), we first establish the causal wage effects of marriage delay. We then propose ways to distinguish among competing theories and hypotheses, as well as the channels through which marriage delay affects wages. Specifically, we take advantage of their different implications for causal relationship, across gender and sub-populations. We reach two conclusions. First, we find a positive causal impact of marriage delay on wages, with a larger effect for women. Comparison of IV and OLS estimates suggests that the *observed* relationship between marriage delay and wages is attributed to both selection in late marriages and true causal effects. Second, we find strong evidence that the positive, causal effects are almost exclusively through increased education for both men and women.

Keywords Timing of first marriage · Wages · Human capital · Selection

JEL Classification J12 · J16 · J31

Responsible editor: Junsen Zhang

✉ Le Wang
Le.Wang.Econ@gmail.com

¹ Department of Economics, The University of Oklahoma,
Norman, OK 73019, USA

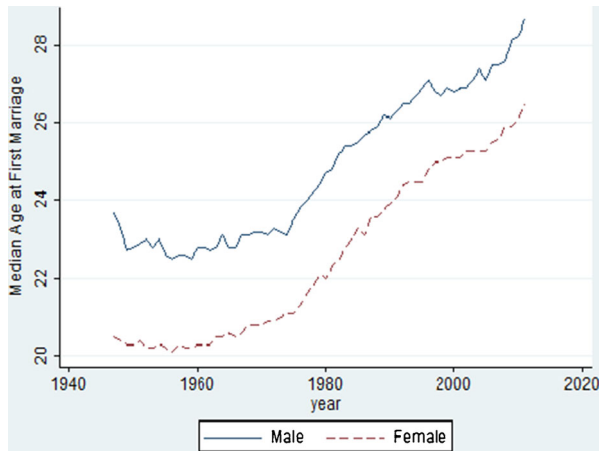


Fig. 1 Median age at first marriage in the USA (1947–2011)

1 Introduction

Over the past few decades, there have been tremendous changes to marriage patterns in the USA, characterized by an increasing postponement of marriage and a sharp increase in age at first marriage. The median age at first marriage in the USA rose from 23.7 in 1947 to 28.7 in 2011 for men, and from 20.5 to 26.5 for women (Fig. 1).¹ This pattern is not unique to the USA and has been similarly observed in many other developed countries (see, e.g., World Fertility Report 2009 by the United Nations). The literature has studied extensively the factors affecting the propensity to marry or the age at first marriage, particularly for women. There have also been numerous studies examining how the delay in first marriage contributes to the changes in overall demographic patterns such as marriage and divorce rates (see, e.g., Loughran 2002 for a thorough review).²

Given that family structures often play an important role in determining individuals' economic outcomes, it is natural to think that the timing of marriage may also affect an individual's labor market performance. Several theories/mechanisms are also put forth to hypothesize the possible relationship between marriage timing and wages. However, very few empirical studies have been carried out to examine the causal wage effects (especially for men)³ and to distinguish among these competing hypotheses. Building on an IV approach proposed in Dahl (2010), we first establish

¹ Source: www.census.gov/population/socdemo/hh-fam/ms2.xls

² In the paper, we use *age at first marriage*, *marital delay*, *delayed marriage*, and *timing of first marriage* interchangeably.

³ The literature has linked marital delay to women's liberation and considered it as a main contributor to increased female labor supply and improved female occupational status (Goldin and Katz 2002; Bailey 2006), but not necessarily wages.

the causal effects of timing of marriage on wages for both men and women. Our major contribution, however, is then to further consider ways to distinguish potential mechanisms that drive the causal relationship, which is largely missing in the literature. Disentangling these different mechanisms would not only help us better understand the wage determination process but it also has important implications for empirical relevance of the competing theories and hypotheses. To achieve this goal, we take advantage of different implications of the competing theories and hypotheses for causal relationship, for men and women, and for different sub-populations. We also address the potential selection issue due to the fact that many women do not participate in the labor market and do not earn any wages.

We focus on some popular mechanisms suggested by several different models and hypotheses. We do want to point out that this paper is not intended to provide an empirical test of every possible mechanism, and that there are alternative mechanisms that we do not consider here. We, however, hope that we have considered some of the important ones, and that the framework provided here could illuminate future work in testing competing hypotheses.

First, Becker (1973) formulates a theory of marriage based on specialization, suggesting that men and women gain from marriage by specializing in home or market work based on their comparative advantages. This theory implies that, on the one hand, if a man gets married earlier, he can fully specialize in the labor market sooner, thereby accumulating more labor market experience and skills and obtaining a higher wage. On the other hand, for women, early marriage interrupts her accumulation of labor market skills or provides an incentive to underinvest in such skills, leading to lower earnings in the labor market. Thus, marital delay should increase the wages of women, but lower the wages of men *ceteris paribus*.⁴ Based on Becker's model, Keeley (1974) similarly argues that high-wage males are more likely to get married at early ages than low-wage males because they can gain more from specialization within marriages.

Second, fertility is an important component of a traditional marriage. To the extent that fertility affects an individual's wages and a marriage precedes child-bearing, delayed marriage could lead to delayed fertility and hence wages. As noted in Lundberg and Rose (2002), fertility can affect an individual's wages through two effects. On the one hand, consistent with Becker's theory, there is the *specialization effect* due to the increased value of a woman's time relative to that of her husband's. As a result, women will focus on home production while men will specialize in the labor market. On the other hand, fertility can also lead to increased value of both parents' time at home on child care, which could have a negative effect on labor market outcomes: the so-called home-intensity effects. For women, both specialization effects and home-intensity effects of fertility on wages are negative, consistent with the motherhood penalty commonly found in the literature. For men, specialization

⁴One may argue that specialization could be achieved via cohabitation and thus does not require legal marriage. This is one of the reasons why we use the 1980 Census data in this paper, as cohabitation was less prevalent during this time period. Further, as noted in Rose (2001), although specialization can be realized through cohabitation, marriage provides for "greater ability to monitor and enforce agreements than more informal relationships" ((Lundberg and Pollak 1995).

effects and home-intensity effects could be opposing, and the net effect of the two is thus theoretically ambiguous. There is however a consensus finding of fatherhood premium, indicating that the specialization effect may prevail (Hersch and Stratton 2000; Lundberg and Rose 2000; 2002). In light of the evidence, we may again find that delayed marriage is positively associated with women's wages, but negatively associated with men's wages. The fertility channel, while important, could operate as the specialization channel as in the basic Becker's model. We thus call this an extended Becker's model.

Third, in addition to its effects on labor market skills, marriage delay may also interrupt the accumulation of formal education. Marriage comes with family responsibilities that often impede the pursuit of formal education for both men and women. Field and Ambrus (2008) find that an additional year of marriage delay leads to an increase in years of schooling by 0.22 for women in Bangladesh. Goldin et al. (2006) attribute the reversal of gender gap in college in the USA to rising age at first marriage, along with changes in expected labor force participation and behavioral problems among boys. Kerckhoff and Parrow (1979) also show that men who marry at younger ages have significantly lower education. As such, marriage delay would be beneficial for both men and women by allowing them to accumulate more formal education. We call it *formal education hypothesis*.

Fourth, Bergstrom and Bagnoli (1993) propose a theory based on asymmetric information, where information on the earnings capabilities of males is only available at later stages of the life cycle. As a result, men with high earnings potential will postpone marriage until their earnings potential is revealed; less capable men will prefer to marry earlier. This suggests that marital delay is positively associated with earnings for men. In addition, this theory suggests that the observed positive relationship is due to self-selection instead of a *causal* effect. Moreover, note that the theoretical result does not depend on any assumption about the timing when the earnings potential is revealed. Similarly, women who marry at different ages may also differ in their preferences. For example, women who are more career-oriented may be more likely to delay marriages (perhaps due to potential interruption of accumulation of both labor market skills and formal education from early marriages), that is, positive selection into late marriages (Korenman and Neumark 1992; Hersch and Stratton 1997). We call this *selection hypothesis*.

Finally, borrowing insights from the job search and migration literature, Loughran and Zissimopoulos (2004) argue that quality job matches (between employers and employees) are in general achieved only after several job changes which often involve migration. Gladden (1999) documents that women tend to sacrifice their career for their spouses more than men do when making migration decisions and indeed finds that women's earnings decrease following a move. This line of reasoning suggests that although marriage may impede worker mobility for both men and women, marital delay will be beneficial for women, but may have smaller effects for men.

These predictions of competing theories and hypotheses are summarized in Table 1. These competing theories provide a way to empirically test the implied mechanisms. Becker (1973) and the extended model, the formal education hypothesis, Loughran and Zissimopoulos (2004), point to a *causal* relationship between age at first marriage and wages (through labor market skills, formal education, or mobility),

Table 1 Theoretical predictions

	Marriage delay and male's wage	Marriage delay and female's wage
Becker's specialization theory	—	+
Extended Becker's model (fertility)	—	+
Formal education hypothesis	+	+
Selection hypothesis	+	+
Loughran and Zissimopoulos' mobility hypothesis	+ or ≈ 0	+

while the selection hypothesis such as Bergstrom and Bagnoli (1993) suggests that the relationship is merely association arising from the unobservable income earning prospects. If we were able to address the endogeneity and omitted variable problems, we could know whether age at first marriage indeed has a causal impact on wages, and hence evaluate the empirical relevance of the models or hypotheses that suggest causal mechanisms as a whole. We can then further empirically distinguish among these causal theories and hypotheses.

Many believe that for women, the timing of marriage and the timing of fertility were usually close, and that the effect of marital delay on wages is potentially mainly through the fertility channel. Indeed, Ellwood and Jencks (2002) document that 75 % of women had their first births within 3 years after their first marriages in 1960. This pattern has, however, changed since then. Ellwood and Jencks (2002) report that in 1990, less than 50 % of women gave birth within 3 years of their first marriages. More recently, “for women as a whole, the median age at first birth (25.7) now falls before the median age at first marriage (26.5).”⁵ As such, it is possible for delayed marriage to have a direct effect on women's career, independent of fertility effects, or through other channels.

Two approaches—panel data and instrumental variable (IV) approaches—are commonly used to control for the endogeneity problems to identify the causal effects. As discussed in more detail below, the panel data approach ignores important sources of the endogeneity problems, and because age at first marriage is time-invariant, identification of its effects using this approach also requires some ingenious tricks.

In the absence of randomized experiments, the only viable approach is, in our view, the IV approach. Moreover, individual characteristics generally are not valid IVs because they may either be the determinants of wages themselves or outcomes that are related to other unobservable characteristics. As such, we have to resort to state-level or aggregate policy changes that could directly impact one's marriage decision. Thus, we borrow such IV used in a similar context for our purpose. In particular, we follow (Dahl 2010) by adopting an identification strategy based on exogenous

⁵Source: New York Times http://douthat.blogs.nytimes.com/2013/03/22/late-marriage-and-its-consequences/?_php=true&_type=blogs&_r=0

variations in minimum marriage age laws across states and over time. Using micro-data from U.S. Census 1980, we reach several important conclusions. First, we find that delayed marriage has a positive effect on wages for both men and women, with the effect being larger for women. Comparison of IV and ordinary least squares (OLS) suggests existence of positive selection into late marriages. Second, the effect varies across age groups. In particular, for both men and women, the effect declines monotonically as one gets older. As we will discuss in details below, these interesting patterns are incompatible with Becker (1973). Further analysis is strongly in favor of the formal education hypothesis—the positive effect of delayed marriage on wages is mostly through increased educational attainment for both men and women. Individuals who delay their first marriages tend to accumulate more formal education, thereby enjoying higher earnings.

Prior to continuing, we want to stress that our purpose in this paper is not to propose a novel IV. This is by no means a trivial task facing any empirical analysis. Dahl (2010) has convincingly argued the validity of the minimum marriage age laws as an instrument for the timing of first marriage. Our focus here is to build on this insight to discuss ways to identify the causal effect of marriage delay on earnings for both men and women, and examine *how* marriage delay may affect wages. This is an important departure from Dahl (2010). The robustness of our results can be readily assessed when alternative identification strategies are available. We believe this practice is standard and necessary for scientific inquiry. Note also that following the literature (surveyed in Section 2.1), we focus on wages, instead of poverty status as in Dahl (2010), because it is more closely related to the labor market outcome in the theories and hypotheses that we are trying to test. These estimates are of interest and speak directly to the average causal per-unit effects of marriage delay on wages.

In addition, our paper contributes to the literature in other ways. The literature has often cited monetary gains as an explanation for the increasing trend of age at first marriage. Our results provide some indirect evidence for this explanation. If such labor market rewards exist, as found in the paper, it is conceivable that individuals may respond to such rewards to change their behavior, which eventually result in a shift in the overall family patterns, especially the increased age at first marriage and the decrease in marriage rates.

Even though our data are drawn from 1980 Census (below, we discuss in greater detail the reasons behind our choice of the 1980 Census), we believe that our results are still general and informative for the following reasons. First, one of the purposes of this paper is to test competing theories. The applicability of theories should not depend on time or a specific population. The fact that our results provide evidence contradicting the existing theories is useful for guiding us to develop a more unified framework to explain the marriage patterns. Second, there has been evidence that skill-biased technical change exists in favor of skilled labor, and as a result, the returns to formal education increase accordingly. Given the uncovered channel through which age at first marriage affects wages, we can only expect the effect to increase over time, and thus our estimates may provide a plausible lower bound for the effect using later datasets. Finally, our results could have potential implications for developing countries during a similar stage of development. Specifically,

these results echo Field and Ambrus (2008)'s finding that enforcing universal age of consent laws could potentially improve women's status in developing countries.

The remainder of the paper is organized as follows. Section 2 reviews some related theoretical and empirical literature. Section 3 presents the empirical method and Section 4 describes the data. Section 5 discusses the results and section 6 provides further discussions of these results. Section 7 addresses the selection issue and Section 8 concludes.

2 Literature review

2.1 On testing competing mechanisms

Empirical findings on the relationship between marital timing and labor market performance are still scarce and mixed; the literature thus provides inconclusive evidence for the empirical relevance of the theories above. Bergstrom and Schoeni (1996) find a positive association between actual age at first marriage and annual earnings for men, but no strong relationship for women. Their paper provides some support for Bergstrom and Bagnoli (1993). Zhang (1995) distinguishes between Becker (1973) and Bergstrom and Bagnoli (1993) by examining the relationship between marital timing and wages separately for men with working wives and men with non-working wives. The idea is that if Becker's specialization theory plays a more important role, the negative relationship should be stronger among men with non-working wives (for whom specialization may be more prevalent). On the other hand, if Bergstrom and Bagnoli (1993)'s theory holds, the positive effect should prevail among men with working wives. The author finds some supporting evidence for both theories.

However, these studies fail to sufficiently address the endogeneity of age at first marriage, and thus do not provide a direct test of these theories. Recent, notable exceptions are Loughran and Zissimpoulos (2004), Field and Ambrus (2008), and Dahl (2010). Loughran and Zissimpoulos (2004) utilize the National Longitudinal Survey of Youth (NLSY) 1979 panel data to estimate a fixed effects model to circumvent the endogeneity problem. There are two potential shortcomings associated with this approach. First, this approach could still fail to address some important sources of the endogeneity problem. This approach can enable meaningful causal inferences only when the selection process is on time-invariant unobservables, or when there is no reverse causality. However, either reverse causality or time-varying unobservables correlated with both age at first marriage and labor market outcomes will invalidate causal conclusions. For example, more career-oriented individuals who tend to delay their marriages may earn more, but their attitudes toward career are not observed and (unlike their earning prospects) may change over time. Second, age at first marriage is also time-invariant, and its effects cannot be identified in this context because all time-invariant variables will be wiped out once the individual-fixed effects are included. As a result, the authors have to resort to an ingenious specification to circumvent this issue. For example, in their specification, age at first marriage is

interacted with years ever married in order to identify the effect. However, the effect identified is only the interactive effect, and the main effect remains un-identified (which is the parameter of main interest in this paper).

Although both Dahl (2010) and Field and Ambrus (2008) employ the IV approach to address the endogeneity of the timing of first marriage, their results provide only a partial test of the existing theories. For example, Dahl (2010) focuses on the impact of early marriage on poverty for women *only*. His finding of a positive impact of early marriage on poverty suggests a positive relationship between late marriage and earnings for women, which could be potentially consistent with all the theories above. Field and Ambrus (2008) find that marriage delay increases the level of educational attainment for women, consistent with the formal education hypothesis. However, whether it has a negative impact on men's human capital accumulation is not the focus of their paper.

In sum, it is important to address the endogeneity issues associated with the timing of first marriage in empirical analysis as well as to analyze the impact separately for men and women. These are the tasks that we hope to accomplish in this paper.

2.2 On potential candidates for IVs

As mentioned above, isolating the causal effect of marriage delay is an important way to distinguish among competing mechanisms. However, this is not trivial because “work and family choices may be undertaken jointly and with foresight” (Miller 2011), and thus, variables affecting one outcome may affect the other as well. In the absence of randomized experiments, a typical solution is to seek the events that mimic the randomized experiments, exogenously changing individuals' marriage decisions but unrelated to individuals' characteristics; it is so-called natural experiment. Related literature has suggested several potential candidates for this purpose. For example, as mentioned above, Field and Ambrus (2008) use the timing of first menarche as an exogenous variation in the earliest date at which girls can marry, and investigate the effects of early marriage on female education in Bangladesh. The literature has also related shocks to marriage markets (i.e., changes in sex ratios) to a decline in the marriageable men available, which in turn affects individuals' marriage decisions. For example, Shemyakina (2007) studies the impact on marriage market of the armed conflict during 1992–1998 in Tajikistan. She finds that war-related deaths decrease the number of marriageable men in the regions affected by the conflicts, and the resulting decline in the sex ratio partially contributes to the decline in marriage hazard for younger women. Charles and Luoh (2007) examine how rising male incarceration decreases the probability of being married for women, and in turn their well-being.

While these events above all seem promising candidates as exogenous shocks to individual marriage decisions to identify the causal effect, they may not suit the analysis in our context. In particular, they all are potentially correlated with unobservable characteristics that determine individuals' wages. For example, the

timing of menarche may be correlated with the family characteristics that influence both adolescent maturation and individuals' long-run labor market prospects. While sibling-fixed effect estimation may help alleviate the problem, siblings tend to live separately after getting married, and thus, it is difficult to obtain information on both adult siblings. Also, lack of variation in the sibling differences in the timing of menarche may impede us from obtaining precise estimates. Moreover, the information on the timing of menarche is generally not available in the large representative Census data and not applicable for men (as we have discussed above, comparison of the effects across gender is important to assess the empirical relevance of various economic theories and hypotheses). On the other hand, war and the rates of male incarceration are not related to individual or family characteristics and largely exogenous to individuals, but they may be related to the characteristics of the regional economy (where individuals work) and hence individuals' wages. And again, these variables are generally not good exogenous shifters in marriage decisions for men, who are an important part of the current analysis. These discussions and considerations help to clarify the issues that we need to pay attention to when finding a potential IV.

The discussions above motivate us to build on Dahl (2010)'s strategy. Specifically, we use exogenous variations in minimum marriage age laws across states and over time. In the USA, it is illegal for youths under certain age to get married, regardless of parental consents, and the age requirement differed substantially across states and over time. The variations in the laws lead to exogenous variations in the legal costs of getting married under the minimum marriage age. Binding legal requirements would shift the timing of marriage for all the individuals in the state and during a certain period of time, which could in turn create norms concerning the age at first marriage. Both legal and cultural constraints could have an impact on the timing of marriage among individuals (shown below). Changes in policies at state level (or more aggregate level) are usually considered to be credible exogenous variations and more likely to be exogenous. As a result, we opt to use these variations in the minimum marriage age laws to identify the effect of the timing of marriage on earnings. Below, we provide more discussions of the IV in the empirical section.

2.3 Summary

In light of discussions regarding the estimation techniques and potential candidates for IVs, we believe that the IV approach is preferred to other approaches in that it sufficiently addresses the endogeneity and measurement error problems. The IV that we opt for—the minimum marriage age laws—also seems to be a straightforward choice in that it directly affects one's age at first marriage. Given this IV, we have to utilize a sample that is most recent but provides enough variations in the laws across states and time for our analysis and hence the 1980 Census (discussed in details below). We believe that the data, sample, and methods used in this paper are well suited and present a first attempt to study the questions at hand.

3 Empirical methodology

3.1 Basic setup

To begin, we estimate the following model:

$$\log(\text{wage})_{i,j} = \beta_{0,j} + \beta_{1,j} \text{age at first marriage}_{i,j} + x'_{i,j} \beta_{2,j} + \epsilon_{i,j}, \quad (3.1)$$

where the outcome variables, $\log(\text{wage})_{i,j}$, pertain to the log hourly wages of an individual i belonging to group $j = \text{male, female}$. This equation is estimated separately for males and females. $\beta_{1,j}$ is the parameter of interest, capturing the returns to marital delay (measured as age at first marriage). $x_{i,j}$ is a vector of (exogenous) individual characters such as age dummies (to capture cohort effects), race dummies, dummy variables for state of *birth*, and regional dummies (see Section 3.2.2 for the discussion of use of birth place).⁶ $\epsilon_{i,j}$ is the error term as usual. Standard errors are clustered at the state level in all estimations.

As most studies do, we exclude various determinants of wages such as educational attainment from the estimation, as these variables are potentially endogenous variables that could be influenced by an individual's decision about her timing of first marriage. That is, these variables themselves could be the reasons why age at first marriage affects individuals' wages. We thus condition on only exogenous variables here to simplify the interpretations of the estimated coefficient—the coefficient captures the *total* effects of age at first marriage at wages. However, we do experiment below with inclusion of several important variables to examine the potential mechanisms through which marital delay affects wages.

Note, also, that marriage year is not a suitable control variable because it is perfectly collinear with age at first marriage and the survey year 1980.

We can consistently estimate $\beta_{1,j}$ via ordinary least squares (OLS) estimation if, conditioning on exogenous characteristics, age at first marriage is uncorrelated with unobservable determinants of wages (or more formally, $\mathbb{E}[\text{age at first marriage} \cdot \epsilon | x] = 0$). However, such assumption may be violated for several reasons. For example, as suggested in Bergstrom and Bagnoli (1993), men with better income prospects may postpone their marriages until their earnings capabilities are fully revealed. Also, more ambitious individuals who could potentially earn more may postpone their marriages in order to pursue their careers. Both examples suggest that $\mathbb{E}[\text{age at first marriage} \cdot \epsilon | x] \neq 0$ and more likely, $\mathbb{E}[\text{age at first marriage} \cdot \epsilon | x] > 0$. As a result, OLS estimates would be biased.

A typical solution is to employ an instrument variable (IV) approach. The IV approach requires existence of an exogenous variable, Z , that is *strongly* correlated with age at first marriage, but uncorrelated with unobservable determinants of an individual's wages. Changes in policies at state level (or more aggregate level) are usually considered to be credible exogenous variations and thus often used as an IV. As such, we adopt the minimum marriage age laws at state level as our excluded instrument. As discussed in the literature review above, given this legal constraint,

⁶We also use age and age squared in place of age dummies, but the results change only slightly.

one is less likely to marry early in a state with more stringent requirement. They are plausibly exogenous to earnings potential, and unlike instrument variables such as individual characteristics (e.g., the timing of menarche), these law changes cannot be manipulated and are not joint decisions with individual work outcomes. As we will show below, there exist enough variations in the laws that would help identify the effect of interest. Given the specialized purpose of these laws, it is less likely that these laws affect individual earnings outcomes through other channels or have any independent effects (unlike the changes in sex ratios discussed above).

To be more precise, we also estimate the following first-stage regression:

$$\text{age at first marriage}_{i,j} = \gamma_{0,j} + \gamma_{1,j} \text{minimum marriage age}_{i,j} + x'_{i,j} \gamma_{2,j} + u_{i,j}, \quad (3.2)$$

where $u_{i,j}$ is the error term as usual. Note that the first-stage equation is nothing but a (parsimonious) linear projection of the endogenous variable on the IV, which does not have to be correctly specified (as long as the linear projection indeed depends on the IV). In this context, the first stage “always involves writing an endogenous variable as a linear projection onto all exogenous variables,” and “there is nothing necessarily structural about” the first stage (Wooldridge 2002, p. 84). The consistency of IV estimates depends only on the exogeneity and strength of the IV, not on the correct specification of the first-stage equation, an insight that goes back to Kelejian (1971) (Angrist and Pischke 2009). This insight is very important because it also allows for measurement errors in the IV due to various reasons (see below for an example of this issue).

3.2 More discussions of the IV validity

There are several potential concerns about our instrument and its construction in practice. Although we borrow this instrument from the literature and some of these issues have been discussed and addressed elsewhere (e.g., Dahl 2010; Blank et al. 2009), we feel it is important to respond to these concerns and provide more discussions here as well.

3.2.1 Binding laws, frequency of law changes, and the first-stage estimates

First, as we will see below, the minimum marriage age (regulated by the state laws with consent) can be as low as 12 for women and 14 for men. The question is whether these laws are indeed binding. If not, the minimum marriage age would have no impact on one's age at first marriage, thereby failing to provide any variations to identify the parameter of interest. Whether or not this would be an issue will be reflected in the coefficient $\gamma_{1,j}$ in our first-stage regression. Dahl (2010) has extensively discussed and examined this issue for women. We further extend this result for men in this context.

Another concern is that changes in the minimum marriage age laws may be small and infrequent and do not differ much across states and time. This is crucial because this could result in a tenuous relationship between the law and one's

marriage decision, even if such relationship exists. It is well known that when instruments are weakly correlated with the endogenous variables, IV estimates are biased toward OLS estimates and inference is not reliable: the so-called weak instruments problem (e.g., Bound et al. 1995; Staiger and Stock 1997). Hence, we conduct several tests to assess the relevance of our instrument. Specifically, we report the first-stage F-statistic for the instrument, as well as Angrist-Pischke first-stage χ^2 underidentification test.

3.2.2 Construction of IV based on state of birth at the time of marriage

To construct the IV, we merge the census data with the marriage age law data by state of *birth*, gender, and year of first marriage. This practice follows (Blank et al. 2009) and matches the individual's state of birth to the state's marriage law since the census data do not provide information on the state of residence at the time of first marriage. Admittedly, this practice is a result of data limitations, but we believe that it is inconsequential in our context for two reasons. First, minimum age laws vary primarily across teenage years, and it is conceivable that the source of identifications comes mainly from teenagers who were impacted by such laws. As noted in Cadena (2013), teens have very low rates of cross-state mobility, and they generally do not make their own living decisions ("independently of their larger household") in response to external changes. As a result, while there may be some cross-borders marriages, the information on state of birth should be relatively more accurate for this group of individuals, and the IV strategy still holds.

Second, state of birth allows us to capture not only binding legal constraint but also general attitude toward the timing of marriage in the environment where one grew up. For example, there has been evidence that peer effects affect individuals' entry into marriage (e.g., Balbo et al. 2013). As a result, minimum marriage age laws in the state of birth should also affect his or her own marriage behavior. More important, even though there may exist mismatched information between one's age at first marriage and the *actual* minimum marriage age law facing the individual then, we should remember that measurement errors in the IV itself do not bias the estimates. As mentioned above, the consistency of IV estimates does not depend on correct specification of the first-stage equation. Measurement errors affect only the efficiency of our estimates, provided that the IV is strongly related to the endogenous variable. This is less of a concern in our context because we show below that the state law is indeed highly correlated with age at first marriage and the effects are precisely estimated.

3.3 Several (innocuous) departures from Dahl (2010)

To examine competing theories, we build our analysis on the IV approach proposed by Dahl (2010). Due to the differences in the specific questions that we are trying to answer, our specifications are slightly different from the one used in Dahl (2010). However, these departures should be largely inconsequential.

First, following most of the literature (surveyed in Section 2.1), we are interested in the age of first marriage, a continuous measure of the timing of marriage, while

Dahl (2010) focuses on early marriage, a binary measure of the timing of marriage. Our focus on a continuous measure is motivated by our interest in the effects of the timing of *actual* marriages. There exists no *a priori* information about the correct specification. However, we know that the linear regression is the best approximation to the potentially nonlinear relationship in terms of mean squared errors. The coefficient, $\beta_{1,j}$, represents the average causal per-unit effects (Angrist and Imbens 1995). Angrist and Imbens (1995) show that when the endogenous variable is incorrectly classified as binary, the sign of the average causal per-unit effects is still consistently estimated, but the resulting estimates are usually larger than the average per-unit effects. Indeed, we find that Dahl (2010)'s estimates are of the same sign as ours, but are larger in terms of magnitude.

Second, note that here we use a continuous IV, instead of discrete dummy variables as in Dahl (2010). This however does not affect the IV results. Indeed, we repeat all of our analysis below using dummy variables as IVs and find the estimates are similar and the patterns remain the same. The results are available from the authors upon request. These results are not surprising because, as mentioned above, the consistency of the IV estimates depends on the exogeneity of the IV, instead of the first-stage specifications.

Finally, we match state laws to the time of first marriage, while Dahl (2010) considers matching the laws to when the person was at age 15. State laws at the time of marriage are most directly relevant for decision making, and it should thus be most statistically related to the endogenous variable, which in turn leads to more precise estimates in both the first- and second-stage estimations. On the other hand, we fail to find that the law in the birth state when the person was age 15 was statistically correlated with his/her marriage. This result is not surprising. The laws when the person was age 15 are exogenous, but it may not *directly* reflect the legal environment at the time when an individual's marriage decision is made. Moreover, for those who were married before age 15, the laws at age 15 are also completely irrelevant. As a result, the first-stage correlation could thus be relatively small. This result is, however, not in contrast to those reported in Dahl (2010), who uses early marriage, a binary variable (as opposed to age at first marriage, a continuous variable), as the dependent variable in the first stage. As noted in (Koenker and Hallock 2001, p. 148), these types of binary outcome models are equivalent to conditional quantile regression models. That is, the effects of minimum marriage age laws on the timing of marriage reflect the effects for those individuals in the lower tail of the marriage age distribution. It is reasonable that the effects of minimum marriage age laws on marriage age are much greater for those in the lower tail of the distribution (those who are likely marry much younger); this can explain why state laws at age 15 may have a more significant effect on a binary outcome than on a continuous outcome.⁷

⁷ As a referee points out, if states tend to raise their minimum marriage ages over time, people who marry at later ages (for reasons unrelated to marriage age laws) will tend to be covered by higher minimum marriage ages than others in their same state and birth year cohort. This may inflate the first-stage estimates using our IV. However, state laws in minimum marriage age during that period did not necessarily follow a monotonic increasing trend; there have been quite a few increases and decreases during the time period, which are important sources of the identification in both our and Dahl's papers.

4 Data

The main data are obtained from the Integrated Public Use Microdata Series (IPUMS) 1980 5 % sample (Ruggles et al. 2010). To perform the analysis, two variables are of primary interest: age at first marriage and hourly wage. We choose the Census 1980 for our main analysis of the impact of marriage delay on individuals' wages, because it contains information on marital history which can be used to calculate retrospective age at first marriage. It also contains information on earnings for a large and representative sample of men and women, weeks worked and hours worked per week, allowing us to calculate the hourly wage. More recent data such as the IPUMS 1990 and 2000 do not contain detailed information on marital history, while the 1970 sample does not have information on weeks worked and hours worked per week. More important, note that since the law requirements have converged universally in the USA, our identification strategy based on minimum marriage age laws is also not available for more recent data.

Our IV strategy follows closely (Dahl 2010). To obtain the instrument variable for age at first marriage, we collect state legal marriage age laws. There are two sets of marriage age laws—age with and without parental consent. A person can get married after a certain age without consent from parents; we follow the literature and call such minimum age the “non-consent” age. The person may also get married at a younger age with the consent of their parents, and such minimum age is called the “consent” age. We follow Dahl (2010) to collect data on consent age laws across states from 1936 to 1969 using sources of *World Almanac and Book of Facts*. These laws could differ for men and women within a state. For women, the consent age can be as low as 12 and ranges to 18, whereas the nonconsent age ranges from 15 to 21. For men, they range from 14 to 21, and 16 to 21, respectively. Over time, there were substantial variations in both the consent and nonconsent ages across and within states needed for identification. We use the consent age as our instrument variable since it defines the lowest possible age that an individual could get married in a state, which is the more binding law.

For a few state-year-gender cells that we cannot find exact information on the law or have no such laws, we treat them as missing and exclude them from analysis. Also, because of the limits on collecting state law information, our analysis will focus on first marriages that occurred between 1936 and 1969 during which our state law information is collected. We then merge the census data with the marriage age data by state of *birth*, gender, and year of first marriage. Year of first marriage is calculated from age at first marriage.

We further restrict the sample to native-born married individuals aged 25 to 60 at the time of the survey and do not live in group quarters. We keep only those observations with positive wage income, weeks worked, and hours worked. The outcome variable of interest is hourly wages instead of annual earnings. As noted in Loughran and Zissimopoulos (2009), using hourly wage as the outcome variable allows us to isolate productivity effects of marital delay from labor supply effects, which is more

consistent with the theories we discussed.⁸ The hourly wage is obtained by dividing total individual annual wages by usual hours worked per week times weeks worked. We define three race groups—white, black, and other, and four regions—Northeast, Midwest, South, and West.

Below we also conduct some preliminary analysis of potential channels through which age at first marriage may affect hourly wages—educational attainment, mobility, and fertility. Educational attainment is measured as years of schooling. We use a (crude) measure of mobility—whether one moved in the past 5 years, which is an indicator equal to one if someone did and zero otherwise. The fertility variable is a dummy variable indicating whether someone ever gave birth, equal to one if yes and zero otherwise. Note that only women were asked about their fertility information, and thus, this information is missing for men. We will discuss these measures further below in Section 6.2.1.

Summary statistics of the sample are provided in Table 2. The average age at first marriage during the time period of our study was 22.74 for males with a standard deviation of 3.91, and 20.27 for females with a standard deviation of 3.57. This is consistent with the observed fact that the median age at first marriage for men is usually two years higher than women, as evident in Fig. 1. Since we are analyzing the sample in the 1980 survey who got married between 1936 and 1969, the average age of our sample is mid-40s. This also allows us to examine the impact of marital delay on individuals' wages later in life.

5 Baseline results

5.1 OLS results

Table 3 presents the OLS estimates of the effect of age at first marriage on wages separately for men and women. Bertrand et al. (2004) show that failure to account for the correlation within the geographic areas where the policy is implemented could severely bias the statistical inference. Dahl (2010) also points out that this issue could be particularly acute in our analysis below when using variations in minimum marriage age laws over time as IV, “because there is typically a long time component and plausible serial correlation.” Thus, standard errors are clustered at the level of state of birth (throughout the paper), which allows for arbitrary correlation over time within a state.

Examining the results in Table 3, we first find that age at first marriage is positively associated with wages for both men and women. This result implies that marital delay

⁸Miller (2013) examines the effect of marriage delay on accumulated wealth, and finds no significant effect using OLS. It would be interesting to see if IV estimation changes the results. However, for the same reason mentioned above as well as lack of information in Census data, we do not examine wealth as an outcome variable here, and will leave it for future research.

Table 2 Summary statistics

	Male		Female	
	Mean	S.D.	Mean	S.D.
	(1)	(2)	(3)	(4)
Log hourly wage	2.14	0.70	1.54	0.66
Age at first marriage	22.74	3.91	20.27	3.57
Age	45.19	8.46	43.76	8.82
Age squared	2113.59	769.60	1992.80	784.83
White	0.92	0.27	0.90	0.30
Black	0.07	0.25	0.09	0.29
Other ethnic group	0.01	0.10	0.01	0.10
Northeast	0.21	0.41	0.21	0.40
Midwest	0.38	0.49	0.38	0.48
South	0.23	0.42	0.24	0.43
West	0.18	0.38	0.18	0.38
Years of schooling	12.64	3.26	12.35	2.48
Moved in the past 5 years (1 if yes)	0.19	0.39	0.18	0.39
No. of Obs	886,515		735,775	

Source: IPUMS 1980 available at www.ipums.org

is beneficial for both men and women. Moreover, the effect is much larger for women than for men. In particular, a year of delayed marriage increases women's wages by about 1.82 %, which is more than four times the effect for men (about 0.4 %), as shown in columns (1) and (4).

This result appears to be consistent with such causal hypotheses as the formal education hypothesis and mobility argument in Loughran and Zissimpoulos (2004) (where, while marriage impedes one's mobility and hence wages for both men and women, such detrimental effect is larger for women because they are usually more constrained by family issues.) This result is also consistent with the selection hypothesis, which suggests a positive relationship between wages and age at first marriage for men and women. However, until we identify the causal effects using IV approach, our result cannot completely distinguish among these hypotheses or theories in practice.

5.2 IV estimates

Even though OLS estimates do not provide a clear answer to whether the effect is causal and which mechanism can explain the effect, they are useful benchmarks, and the comparison of OLS and IV estimates would allow us to differentiate *correlation* and *causality* and hence between the selection hypothesis and other causal mechanisms.

Table 3 OLS results: the effect of age at marriage on wages

	Male		Female			
	Full sample	Marriage age ≤ 17	Marriage age [18, 20]	Full sample	Marriage age ≤ 17	Marriage age [18, 20]
	(1)	(2)	(3)	(4)	(5)	(6)
Age at first marriage	0.0038*** (0.0006)	0.0274*** (0.0053)	0.0246*** (0.0025)	0.0182*** (0.0003)	0.0247*** (0.0030)	0.0347*** (0.0015)
No. of Obs	886,515	26,907	232,516	735,775	129,288	319,724

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and birth place dummies

We now turn to the IV results in Table 4. Prior to continuing, we first discuss our first-stage results (panel A of Table 4). As discussed above, we expect that a more stringent minimum marriage age law leads to an increase in age at first marriage. Specifically, we find that an increase of 1 year in the minimum marriage age law increases more than 6 months in age at first marriage (the estimated coefficient is 0.5951 for men and 0.5154 for women and statistically significant at $p \leq 0.01$ levels). The fact that these coefficients are precisely estimated indicates (1) that we do have enough variations in the minimum marriage age laws to identify the parameter of interest and (2) that the laws are indeed binding.

However, as now is well known, test of significance in the presence of weak instrument variables could have incorrect size and lead to misleading inference (Bound et al. 1995). Hence, we also report both F-statistic and Angrist and Pischke (2009)'s (AP) χ^2 test to assess the strength of our IV (panel B of Table 4). The F-statistics for both men and women are larger than the rule-of-thumb threshold value of 10. Moreover, AP χ^2 test also indicates that weak IV is not a concern in this context. In sum, our IV fares well in terms of these diagnostic tests.

Panel C of Table 4 presents the actual IV estimates. We continue to find that marital delay has a positive effect on wages for men and women. In comparison

Table 4 IV results: the effect of age at marriage on wages

	Male (1)	Female (2)
Panel A: first-stage results		
Effect of minimum marriage Age law	0.5951*** (0.1675)	0.5154*** (0.1378)
Panel B: identification tests		
<i>F</i> -test	12.622	13.993
<i>p</i> value	(0.001)	(0.001)
AP χ^2 test	12.886	14.286
<i>p</i> value	(0.000)	(0.000)
Panel C: IV estimates		
Age at first marriage	0.0082*** (0.0027)	0.0177*** (0.0040)
No. of Obs	886,515	735,775

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and birth place dummies

to OLS estimates, the IV estimates of the positive effect doubles for men, while it decreases slightly for women. Moreover, the effect is again larger for women. In particular, the estimated coefficient is now 0.0082 for men and 0.0177 for women.

The remaining question is: whether or not marriage laws is indeed a good instrument and exogenous. One common concern when using policy changes as an IV is whether such changes coincide with other law changes that could have a direct effect on the outcome. For example, marriage laws could be correlated with compulsory schooling laws, violating the exogeneity assumption of a valid IV. To this end, we repeat our analysis by controlling for compulsory schooling laws in two ways. First, we use compulsory schooling laws as a control variable (by controlling for compulsory schooling laws in both the first- and second-stage estimations). Second, we use compulsory schooling laws as an IV (by controlling for it only in the first-stage estimation.) As we can see in Table 5, our IV results continue to hold, and the differences in the estimates before and after including compulsory schooling laws are statistically insignificant.

5.3 Comparison of IV and OLS estimates: selection vs. causal mechanisms

As mentioned above, comparison of OLS and IV estimates can be informative of the direction of the selection bias and thus serve as a useful test of the selection hypothesis. The comparison of full-sample OLS and IV estimates, however, may not necessarily be a fair evaluation of the hypothesis. As pointed out in Imbens and Angrist (1994), the IV estimates capture only the effect for a subgroup of individuals who are most likely impacted by the IV (i.e., the compliers). In our context, it is conceivable that the minimum marriage age law affects mainly teenagers. To this end, we re-estimate the OLS models for men and women who got married at less than 17 years old, and between 18 and 20 (columns (2), (3), (5), and (6) of Table 3), and find that the OLS estimates are larger for the younger groups than those for the full sample, and that they are larger than the IV estimates.

These results support the positive selection hypothesis. Remember that the selection hypothesis suggests that the positive association (OLS estimates) among men arises from the unobservable earnings prospect—men who delayed their marriage are usually those who have better earnings potential and indeed earn higher wages later (i.e., a positive selection into late marriages). This implies that OLS estimates are biased upward. Addressing the endogeneity issue indeed leads to smaller IV coefficients. Similar reasoning applies to women.

Since the IV estimates continue to be positive and statistically significant, there exist a *causal* relationship between marriage delay and wages. The next question is whether we could separate the causal mechanisms.

6 Investigating the causal mechanisms

The sign and magnitude of the IV estimates seem to be consistent with the formal education hypothesis and the mobility hypothesis in Loughran and Zissimopoulos

Table 5 IV results: the effect of age at marriage on wages controlling for compulsory schooling laws

	Compulsory schooling law as a control		as an IV	
	Male (1)	Female (2)	Male (3)	Female (4)
Panel A: first-stage results				
Effect of minimum marriage	0.5414***	0.4795***	0.5414***	0.4795***
Age law	(0.159)	(0.141)	(0.159)	(0.141)
Panel B: identification tests				
<i>F</i> -test	11.6422	11.6186	21.5787	25.6083
<i>p</i> value	(0.001)	(0.001)	(0.000)	(0.000)
AP chi ² test	11.886	11.8621	88.122	104.5803
<i>p</i> value	(0.001)	(0.001)	(0.000)	(0.000)
Panel C: IV estimates				
Age at first marriage	0.0079*** (0.0025)	0.0164*** (0.0038)	0.0052*** (0.0020)	0.0194*** (0.0029)
No. of Obs	883,582	733,138	883,582	733,138

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and birth place dummies

Minimum compulsory attendance is controlled for in IV estimations. In columns (1) and (2), it is used a control variable, appearing in both first- and second-stage estimations. In columns (3) and (4), it is used an IV, appearing only in first-stage estimations. The data are obtained from Acemoglu and Angrist (2000)

(2004), but inconsistent with Becker (1973) and its extended model. However, as suggested by Zhang (1995), it is possible that two competing theories may be at work at the same time. For example, recall that Becker's theory predicts a negative effect of delayed marriage on men's wages and a positive effect on women's wages. It is possible that the negative effect for men is smaller than the positive mobility and education effect, and hence, we observe a net positive effect for men. And because both theories suggest a positive effect for women, it is not surprising that the net effect for women remains positive.

In what follows, we discuss several ways to differentiate between various causal mechanisms. Prior to continuing, we want to point out that different approaches rely

on different assumptions; however, our conclusion does not depend on a specific approach. Instead, all results seem to be strongly in favor of the formal education hypothesis.

6.1 Heterogeneous effects across age groups: an evaluation of various causal mechanisms

Our first attempt relies on Zhang (1995)'s insight that competing theories, if they co-exist, may manifest themselves differently for different sub-populations. The effect suggested by one theory may be stronger for one group, while the effect by another theory may prevail among another group.⁹ Therefore, examining heterogeneous effects across sub-populations can provide a useful means to distinguish among theories.

Here, we examine the heterogeneous effects across age groups for both men and women by repeating our IV estimations separately by age groups. This approach is motivated by two considerations. First, Loughran and Zissimopoulos (2004)'s mobility hypothesis would predict no variations across age groups. Once the level of job match is achieved, it is reasonable to assume that the match effect remains constant (Low et al. 2010). As a result, the positive effect arising from the mobility effect should similarly remain constant across age groups as well.

In contrast, the formal education hypothesis and Becker's specialization theory suggest that the effects of marriage delay should decrease in magnitudes with age. This is because both hypotheses involve accumulation of some form of human capital, either formal education or on-the-job training, which can deteriorate and even become obsolete over time and with age. Even though few estimates exist of how depreciation varies over time, it is "intuitively plausible to hypothesize that depreciation of skills probably increases with age" (Polachek and Siebert 1993, p. 33). Supporting this hypothesis, Henderson et al. (2011) find that returns to education decrease monotonically with age groups. This implies that the human capital mechanism may become less important as one gets older. As a result, Becker (1973) implies that the negative effect of delayed marriage for men should decrease (in absolute terms) as one gets older. The positive effect for women should also similarly decrease. On the other hand, the formal education hypothesis implies that the positive effects of delayed marriage would decrease when one gets older for *both* men and women.

Our results in Table 6 show that there exists a monotonically decreasing trend of the positive effects for both men and women. This result is consistent with the formal education hypothesis, but inconsistent with the implications of both Becker's and Loughran and Zissimopoulos' hypotheses.

An implicit assumption in our analysis is that cohorts are similar over time. Cohort effects may affect the interpretation of the results. Changes in the patterns of

⁹Both Zhang (1995) and Bergstrom and Schoeni (1996) provide some evidence of such patterns. In particular, Bergstrom and Schoeni (1996) find that while age at first marriage is positively associated with wages (consistent with Bergstrom and Bagnoli (1993)), the relationship becomes negative for those who married after age 30 (which cannot be explained by Bergstrom and Bagnoli (1993)).

Table 6 IV estimates of the effects of age at first marriage on wages over the life cycle

	25–30 (1)	31–40 (2)	41–50 (3)	51–60 (4)
Panel A: male				
Age at first marriage	0.0589*** (0.0247)	0.0179*** (0.0042)	0.0066*** (0.0028)	0.0020 (0.0035)
No. of Obs	14,104	291,326	302,100	278,985
Panel B: female				
Age at first marriage	0.0465*** (0.0150)	0.0315*** (0.0043)	0.0182*** (0.0024)	0.0073*** (0.0044)
No. of Obs	32,116	268,907	235,427	199,325

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and birth place dummies

assortative mating over time is an example of such cohort effects, and there has been some evidence of increasing positive assortative mating (i.e., couples have become more similar over time). Assortative mating can, however, be considered as a feature of the original Becker's model. For example, Lam (1988) shows that negative assortative mating stems from specialization in Becker's original model. As noted in Rose (2001), increasingly positive assortative mating is a result of "a decline in the scope for specialization and exchange as women have become more 'like men' in terms of their labor market behavior." This result implies that for more recent, younger cohort, there is less specialization within marriages, and in turn that marriage delay is less beneficial for younger women. This implied pattern is inconsistent with what we find, and thus, assortative mating itself may not necessarily account for our results.

There are other examples of cohort effects such as changes in the labor market for women or the changes in the composition of the compliers when the law changes over time. Note, however, that neither can necessarily account for the observed patterns. For example, while the labor market for women changed dramatically over the relevant cohorts, our tests of the theories and hypotheses rely on *both* men *and* women. For instance, both Becker's theory and the mobility hypothesis are not consistent with the observed pattern for men. This actually underscores the potential usefulness of our approach, which considers the implications of competing theories and hypotheses not only for one particular sub-population, but simultaneously for different sub-populations.

Also, suppose that the compliers are always “marginal” individuals whose marriage decisions are made around the minimum marriage ages set by the laws. Then, the general upward trend in the laws implies that the compliers for the younger cohort may get married later than those for the older cohort. If this is true, then for the compliers among the younger cohort, the effects of formal education are less important, while Becker’s theory and the mobility hypothesis, if they do exist, may become more relevant. This is again not what we find.

There may be other examples of cohort effects that we have not considered here. To this end, instead of *indirectly* testing the implications of the mechanisms, we provide further direct tests of the formal education hypothesis below.

6.2 Further assessment of the formal education hypothesis

In this section, we use three approaches to examine whether formal education is the main channel through which age at first marriage could affect wages. All approaches build on the idea that marital delay should not have any direct effects on wages once the effects of the main channels are (correctly) controlled for in the estimation.

The first approach achieves this by controlling directly for variables capturing varying channels. This approach assumes that such channel variables are exogenous and hence is only preliminary and suggestive. Instead of assuming exogeneity of these channel variables, the second approach borrows from the literature credible, consistent estimates of the main channel variable in the wage equation, and thus, we are able to control for the true effects of this variable in estimations. We focus on education, which as will be shown below, is indeed the main channel. Finally, we also directly instrument for the endogenous education variable using commonly used instrumental variables.

6.2.1 First approach

We now turn to our first approach. This approach proceeds with two steps. The first step is to examine whether age at first marriage has any impacts on the variables capturing potential channels. Second, if a potential channel variable is indeed affected by age at first marriage, we then include this candidate variable in the original wage equation. The idea is that for a variable to be considered as a potential candidate, age at first marriage should actually have a *causal* effect on this variable. Further, if such variable is indeed an important channel, when it is included in the wage equation, the magnitude of the effect of age at first marriage on wages should decrease. Although this exercise is only *preliminary* (because the channels are endogenous and its coefficient is not necessarily consistently estimated), we believe it is still useful and informative.

We first consider two possible channels for both men and women—mobility (measured by whether one moved in the past 5 years) and formal education (measured by education). In addition, we also consider one additional channel for women: fertility

(measured by ever had any children). To estimate the causal effects of age at first marriage on these candidates, we re-estimate our outcome equation that used them as dependent variables—the variables capturing intermediate outcomes (i.e., various channels), instead of the final outcome (i.e., wages), while still instrumenting for age at first marriage:

$$\text{channel}_{i,j} = \beta_{0,j} + \beta_{1,j} \text{age at first marriage}_{i,j} + x'_{i,j} \beta_{2,j} + \epsilon_{i,j}, \quad (6.1)$$

where $\text{channel}_{i,j}$ is a particular intermediate outcome—mobility measure, education, or fertility. The results are presented in Table 7. We find that mobility is indeed affected by age at first marriage. For both men and women, there is a negative effect. The effect is roughly the same for women and men, inconsistent with Loughran and Zissimopoulos (2004). However, recall that the mobility is measured by whether someone moved in the past 5 years. This is only a crude measure of mobility, and we should be cautious about the interpretation of this result.¹⁰

We also observe that formal education is affected by the timing of first marriage. In particular, the effect on education is positive for both men and women. The pattern is consistent with the formal education hypothesis. For women, we show that delayed marriage reduces their probability of having children. Ideally, we would like to use a more refined measure of fertility, such as age at first birth, but such information is not provided in the Census data. This result lends some support for the extended Becker's model which emphasizes the fertility channel.

We now turn to our second-stage results to examine how much of the positive effect previously found can actually be explained by these different channels. The results are reported in Table 8. Because fertility variable is only available for women, we first add fertility into the equation for women (column 3). As we can see, the positive effect decreases by 2 percentage points. This indicates that one of the reasons behind the beneficial effects of delayed marriage is associated with delayed motherhood. This result is not surprising, given the importance of motherhood penalty found in the literature, and it lends some support for the extended Becker's model. However, the fertility channel can explain only roughly 11 % ($= \frac{|0.0157 - 0.0177|}{0.0177} \times 100$) of the positive effect. While the fertility channel is one possible channel through which age at first marriage affects wages, its role is not as big as people generally believe it to be. Consistent with our previous discussions, this result implies that age at first marriage could have an independent effect or through other channels.¹¹

¹⁰ An alternative measure of mobility is interstate move by comparing between state of birth and state of current residence. This is not necessarily a better measure, however. First, such mobility measure also captures involuntary moves during childhood. Second, across-state mobility fails to account for migrations within state. We can use this measure as another crude measure of mobility to test our results. However, as shown below, we do find strong evidence that all the coefficients are completely explained by education, which leaves little room for other channels.

¹¹ One concern is that determinants of child-bearing are also those of marriage, in which case a model for fertility and a model for marriage would be observationally equivalent and cannot be separately identified. This implies that conditioning on fertility, there would not be any independent variations in marriages or age at first marriages. Our results, however, do not support this.

Table 7 IV estimates of the effects of age at first marriage on various channels

	Moved (1)	Schooling (2)	Ever had children (3)
Panel A: male			
Age at first marriage	-0.0041** (0.0016)	0.1468*** (0.0301)	
No. of Obs	883,582	883,582	
Panel B: female			
Age at first marriage	-0.0038** (0.0016)	0.2035*** (0.0384)	-0.0153*** (0.0030)
No. of Obs	733,138	733,138	733,138

Reported in parentheses are robust standard errors clustered at the level of state of birth. Control variables include cohort fixe effects, white, black, region dummies, and birth place dummies

We subsequently add mobility into the wage equation. The coefficients for both men and women remain virtually unchanged. The result suggests that mobility argument may play a rather limited role. The last variable that we add to the model is years of schooling. The result is striking. Specifically, we find that the positive effects completely disappear and become close to zero and statistically insignificant. This result holds for both men and women. This result alone suggests that the most important channel is through the effect on accumulation of formal education.

One may argue that mobility and fertility are crude measures, but the fact that all the coefficients are completely explained by education still underscores the importance of the effect through increased formal education.

6.2.2 Second approach

As mentioned above, another concern of the first approach is that the interpretation hinges on the assumption that returns to education are correctly estimated. To further investigate the importance of the formal education hypothesis, we impose the *true* returns to education in the wage equation and re-estimate our model. Since the effect of education is correctly controlled for, we should expect the effect of age at first marriage to become much smaller or close to zero (the latter holds if the education channel is the only mechanism). The question is how we can obtain the true returns to education. To achieve our goal, we borrow the IV estimates of the returns to education obtained in the influential study in this field by Angrist and Krueger (1991) (where the authors also use the 1980 Census data). Their estimates differ across age groups, and hence we use both the lower (0.06, Column 8 of Table V) and upper bounds (0.0779, Column 8 of Table VI) in our exercise. We also use these estimates for

Table 8 Investigation of different channels (IV estimates)

	Male		Female	
	Add moved (1)	Add moved and schooling (2)	Add fertility (3)	Add moved and fertility Add schooling, moved and fertility (5)
Age at first marriage	0.0082*** (0.0030)	-0.0001 (0.0020)	0.0157*** (0.0040)	0.0006 (0.0030)
No. of Obs	883,582	883,582	733,138	733,138

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and *birth* place dummies

women. The literature generally finds that the returns to education among women are larger than the returns among men (e.g., Dougherty 2005). We thus can regard these estimates as lower bounds for women. The results are reported in Table 9. The positive effect of delayed marriage completely disappears and becomes statistically insignificant once we control for the education channel. This holds for both men and women. This again highlights the fact that formal education is the main channel through which age at first marriage affects wages, consistent with our findings above. Note that one should not take the literal value of the estimates and simply interpret the reduction in the estimates as more than 100 % of the effects without considering the statistical significance.

6.2.3 Third approach: instrumenting for education

Instead of borrowing the estimates from the literature, we can also obtain the consistent estimates for education in our own analysis by instrumenting for education. This is our third approach. The instrument used here is compulsory schooling laws, which should affect individuals' educational choices, but are independent of (or directly related to) their wages. Individuals who are most affected by such laws are those at the margin of completing high school education, and we therefore use high school dropouts to measure formal education in our estimations. The results are presented in Columns (5) and (6) of Table 9. We first notice that high school dropouts fare considerably worse than those who graduated from high school. The wage penalty of high

Table 9 Investigation of the importance of education channel (IV estimates)

	Fixing returns to education				Instrumenting for education (high school dropout)	
	Male		Female		Male	Female
	Lower bound (1)	Upper bound (2)	Lower bound (3)	Upper bound (4)	(5)	(6)
Age at first marriage	0.000 (0.0022)	−0.003 (0.0020)	0.006* (0.0028)	0.002 (0.0027)	−0.0034 (0.0027)	0.0067 (0.0071)
High school dropout					−0.6762*** (0.2032)	−0.4381* (0.2727)
No. of Obs	883,582	883,582	734,124	734,124	883,582	733,138

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and birth place dummies

Columns (1) and (3) assume that returns to education is 0.06, while columns (2) and (4) assume that returns to education is 0.0779. These numbers are obtained from Table V, column 8 and Table VI, column (8) in (Angrist and Krueger 1991), respectively

Columns (5) and (6) use minimum compulsory schooling years as an IV for high school dropout

school drop out is as large as 67 percent for men and 43 percent for women. We find that once education is correctly controlled for in our estimations, the coefficients on age at first marriage decrease drastically and become statistically insignificant. This result again implies that education accounts for nearly all the effects of marriage delay on wages. Another interesting finding is that the direct effects of age at first marriages are surprisingly similar to those obtained using the second approach (with fixed returns to education from the literature).

6.3 Summary

We have thus far taken different approaches to assess the validity of possible underlying mechanisms. Although turning on different assumptions, all approaches seem to suggest that formal education accumulation plays an essential role in explaining the causal effects of marriage delay on wages. As a result, although future studies investigating other channels using our approach would certainly be useful, we believe that our main conclusion would remain unchanged.

7 Role of selection

We have thus far considered only working women in our empirical analysis. Women have relatively low rates of employment and labor force participation, and we do not observe wages for these nonworkers. Further, women's participation in the labor market may vary at different points in their life. To the extent that non-working women systematically differ from working women, our analysis could be biased. This issue could be particularly relevant in our context. For example, the strong version of the Becker's theory of specialization implies that women could drop out of the labor market. Ignoring this could potentially lead to underestimation of the beneficial effects of delayed marriage for women, thereby leading to refutation of the Becker's theory. While we find statistically significant, positive effects of delayed marriage, which could be considered lower bounds of the true effects, it is nevertheless important to examine whether the observed results and patterns are robust to addressing the selection.

7.1 Selection models and validity of exclusion restriction

To address the selection issue, consider the extended system of Eqs. 3.1 and 3.2 in the presence of endogeneity.

$$\begin{aligned}\log(\text{wage})_{i,j} &= \beta_{0,j} + \beta_{1,j} \text{ age at first marriage}_{i,j} + x'_{i,j} \beta_{2,j} + \epsilon_{i,j} \\ \text{age at first marriage}_{i,j} &= \gamma_{0,j} + \gamma_{1,j} \text{ minimum marriage age}_{i,j} + x'_{i,j} \gamma_{2,j} + u_{i,j} \\ S_{i,j} &= I(z_{i,j} \delta - \eta_{i,j} \geq 0),\end{aligned}\quad (7.1)$$

where S is an indicator and equal to one if one participates in the labor market and zero otherwise; and $\tilde{z}$ is a vector of exogenous characteristics, which *can* include a variable not in the set of x . Equation 7.1 indicates that a woman decides to participate

Table 10 Validity tests of spousal income in selection models

	Marginal effects (1)	Validity tests (2)
Spousal income	−0.087*** (0.001)	−10.389 <i>p</i> value = 1.000

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and birth place dummies

The validity test of the IV is developed in Huber and Mellace (2011). The null hypothesis is that the IV is valid

in the labor market when $\tilde{z}\delta - \eta \geq 0$. This equation is consistent with the traditional Roy model. The model can be identified under typical assumptions for IV and selection models and estimated via a variant of the conventional Heckman model (see, e.g., (Wooldridge 2010, p. 809) for details).

As is well known, Heckman type of selection models perform relatively poor even though identification can be achieved through distributional assumption without an exclusion restriction (i.e., $z = x$).¹² To this end, we therefore include an exclusion restriction—spousal income ($\tilde{z}$)—to aid identification; specifically, $\tilde{z}$ equals one if spousal income is greater than median income and zero otherwise. This choice of exclusion restrictions for the female labor supply equation is a popular one in the literature, and similar variables have been used in the previous literature (e.g., Buchinsky 2001; Chang 2011; Martins 2001). Below, we present our (statistical) evidence supporting this choice.

To assess the validity of our exclusion restriction, we present two sets of results in Table 10. The first set is concerned with the *strength* of empirical relationship between our exclusion restriction and labor force participation decision. The literature has generally found strong evidence that spousal income influences a woman's decision to participate in the labor market (e.g., Mroz 1987; Zabel 1993). We present the marginal effects of spousal income on a woman's probabilities of labor force participation. Consistent with the literature, our first-stage results show that spousal income indeed has a negative and statistically significant effect on labor force participation rates among women. Specifically, having a spouse who earns more than median income can reduce female labor force participation rate by roughly 8.7 % (column 1 of Table 10).

The second set of results is concerned with the independence (or exogeneity) of an exclusion restriction; the exclusion restriction must be independent of potential wages (or conditional on X). Such assumption may be violated if spousal income has any direct effects on women's wages, or is indirectly related with women's wages through other channels. One possibility, for instance, is the presence of assortative mating (positively or negatively) on unobservable determinants of individual

¹²The reason is because the inverse Mills ratio will be a function of only x and the reduced form could suffer from collinearity.

income/wages, which implies potential correlation between spousal income and the error term as well. To formally test whether this assumption (along with the monotonicity assumption) is violated, we undertake a formal test based on a novel method recently proposed in Huber and Mellace (2014). They show that under our model assumptions, the following inequalities hold:

$$\begin{aligned}\mathbb{E}[\log(\text{wage})|\tilde{z}=1, S=1, \log(\text{wage}) \leq y_q] &\leq \mathbb{E}[\log(\text{wage})|\tilde{z}=0, S=1] \\ &\leq \mathbb{E}[\log(\text{wage})|\tilde{z}=1, S=1, \log(\text{wage}) \\ &\geq y_{1-q}]\end{aligned}$$

Such inequalities imply the following null hypotheses:

$$\begin{aligned}\mathbb{E}[\log(\text{wage})|\tilde{z}=1, S=1, \log(\text{wage}) \leq y_q] - \mathbb{E}[\log(\text{wage})|\tilde{z}=0, S=1] &\leq 0 \\ \mathbb{E}[\log(\text{wage})|\tilde{z}=0, S=1] - \mathbb{E}[\log(\text{wage})|\tilde{z}=1, S=1, \log(\text{wage}) \geq y_{1-q}] &\leq 0\end{aligned}$$

Huber and Mellace (2014) propose a test procedure to verify these inequalities.¹³ A negative test statistic with a large p value indicates that the IV validity is *not* violated and cannot be rejected statistically. The results are presented in column (2) of Table 10. We fail to reject the validity of our exclusion restriction, strongly in favor of the use of the presence of spousal income as an exclusion restriction for the selection equation. These results, while not necessarily definitive, do increase our confidence in the identification assumption used in our analysis.

7.2 Results

Having discussed the validity of our approach, we now turn to actual estimates addressing the selection issue. The results are presented in Table 11. We repeat all of our analysis addressing the selection. As we can see, all of our results continue to hold. Not only do we find similar patterns in our estimates; we generally find estimates in remarkably similar magnitudes as well. For example, we again find a statistically significant and positive effect of delayed marriage on wages among women. We also find that the effects decline monotonically across cohorts. Most importantly, we continue to find that the educational attainment channel may account for nearly all the positive effects that we observe, regardless of the approaches used.

Two additional results are worth mentioning. First, consistent with our discussions above, the estimates addressing the selection are generally larger than those without addressing the selection. The difference is, however, not statistically significant, implying that the selection issue exist but does not severely bias our results. Second, addressing the selection has varying levels of impacts on the estimates across cohorts. Specifically, the largest impact occurs for the estimate for the youngest cohort. This result indicates that working women may be a more selective group among younger women.

¹³Note that this test can be readily extended to the multivalued case, but for ease of exposition and computation, we consider a binary case here, i.e., whether spousal income is greater than or equal to median income.

Table 11 IV estimates addressing the selection issue

	Full sample	Life-cycle effects				Different channels			Fixed returns to education		Instrumenting for education
		25–30	31–40	41–50	51–60	Fertility	Moved + Fertility	Schooling + Moved + Fertility	Lower bound	Upper bound	
(1)		(2)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Age at first marriage	0.0187*** (0.0041)	0.0959*** (0.0365)	0.0299*** (0.0041)	0.0201*** (0.0025)	0.0084*** (0.0037)	0.0166*** (0.0045)	0.0166*** (0.0045)	0.0016 (0.0038)	0.0062* (0.0032)	0.0024 (0.0032)	0.0068 (0.0066)

Reported in parentheses are robust standard errors clustered at the level of state of birth *** $p < 0.01$, ** $p < 0.05$, * $p < 0.01$. Control variables include cohort-fixed effects, white, black, region dummies, and *birth* place

Columns (10) assumes that returns to education is 0.06, while columns (11) assumes returns to education is 0.0779. These numbers are obtained from Table V, column 8 and Table VI, column (8) in , respectively, dummies

Columns (12) uses minimum compulsory schooling years as an IV for high school dropout

8 Further discussions

Having obtained all the results, it is useful to step back to think about whether these results indeed make sense. To put our results in perspective, consider first that Miller (2011) find that motherhood delay is associated with “a substantial increase in earnings of 9 % per year of delay, an increase in wages of 3 %.” Raising a child is usually considered to require more sacrifices from women (especially in terms their careers). It also likely requires more time from women and imposes a larger burden on them, compared to men. It is thus only reasonable to consider that the effect of motherhood delay provides a useful upper bound for the effect of delayed marriage. And we indeed find that the effect of delayed marriage is much smaller than the effect of motherhood delay. Thus, this is a sensible estimate for women.

Moreover, recall that our preliminary analysis of the underlying mechanisms indicates that the main reason for the beneficial effect of marital delay is due to increased accumulation of formal education. As men generally have higher level of formal education than women, it is then plausible that men’s rewards for delaying marriage are much smaller than women’s. It is thus not surprising that we generally find that the estimates are larger for women than those for men, and that even for women in their 50s, the effect is still significant, while the effect for their male counterparts becomes insignificant and close to zero.

It is also useful to think about the implications of our results. As discussed earlier, IV estimates capture only local average treatment effects, and in our case, likely the teenagers. For this group of individuals, the effect of delayed marriage on wages is more likely through the effect on formal education. It would be interesting to assess whether this channel is still important for the rest of the population when alternative IVs are available. Future research is thus warranted to assess the generality of our results. Even so, our finding is of paramount policy relevance, especially the fact that teenage marriage affects future earnings mainly through educational attainment, and that this applies for both men and women.

9 Conclusions

In this paper, we examine the effect of age at first marriage on wages for both men and women. To isolate the causal effect, we exploit the variations in minimum marriage age laws across states and over time. We find that age at first marriage has a positive effect on wages for both men and women, with the effect being larger for women. Our IV estimates are smaller than OLS counterparts, suggesting the existence of positive selection. We also find that the effect varies across age groups. In particular, for both men and women, the effect declines monotonically as one gets older. The results seem to be incompatible with Becker (1973) and Loughran and Zissimopoulos (2004), but consistent with the formal education hypothesis. Further careful investigation confirms this result. We find that nearly all the beneficial effect of delayed marriage can be attributed to the increase in the level of education resulted from delayed marriage.

Acknowledgments The authors thank Junsen Zhang (the editor) and three knowledgeable referees for their constructive comments. The authors are also particularly grateful to Reagan Baughman, Daniel Henderson, Karen Smith Conway, Amitabh Chandra, Bruce Elmslie, Ju-Chin Huang, Delia Furtado, Sanders Korenman, Ted Joyce, Scott Drewianka, Per Fredriksson, Paul Glewwe, Esfandiar Maasoumi, Daniel Millimet, Solomon Polachek, Subal Kumbhakar, and Christopher Hanes, and seminar participants at the Southern Methodist University, University of New Hampshire, SUNY-Binghamton, University of Nevada-Las Vegas, and SEA Annual Meetings for their invaluable comments and suggestions. Mica Kurtz provided excellent research assistance in collecting the data on minimum marriage age laws. The usual disclaimers apply.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

- Angrist JD, Imbens GW (1995) Two-stage least square estimation of average causal effects in models with variable treatment intensity. *J Am Stat Assoc* 90:431–442
- Angrist J, Krueger AB (1991) Do compulsory school attendance affect schooling and earnings. *Q J Econ* 106:979–1014
- Angrist JD, Pischke JS (2009) Mostly harmless econometrics: an empiricist's companion. Princeton University Press, NJ
- Bailey MJ (2006) More power to the pill: the impact of contraceptive freedom on women's life cycle labor supply. *Q J Econ* 121:289–320
- Balbo N, Barban N, Mills M (2013) Friend and peer effects on entry into marriage and parenthood. A multiprocess approach. Dondena Working Papers
- Becker GS (1973) A theory of marriage: part I. *J Polit Econ* 81:813–846
- Bergstrom T, Bagnoli M (1993) Courtship as a waiting game. *J Polit Econ* 101:185–202
- Bergstrom T, Schoeni RF (1996) Income prospects and age-at-marriage. *J Popul Econ* 9:115–130
- Bertrand M, Duflo E, Mullainathan S (2004) How much should we trust differences-in-differences estimates? *Q J Econ* 119:249–275
- Blank RM, Charles KK, Sallee JM (2009) A cautionary tale about the use of administrative data: evidence from age of marriage laws. *American Econ J: Appl Econ* 1:128–149
- Bound J, Jaeger DA, Baker RM (1995) Problems with instrumental variables estimation when the correlation between the instruments and the endogenous explanatory variable is weak. *J Am Stat Assoc* 90:443–450
- Buchinsky M (2001) Quantile regression with sample selection: estimating women's return to education in the U.S. *Empir Econ* 26:87–113
- Cadena BC (2013) Recent immigrants as labor market arbitrageurs. Evidence from the minimum wage. Unpublished Manuscript
- Chang SK (2011) Simulation estimation of two-tiered dynamic panel Tobit models with an application to the labor supply of married women. *J Appl Econom* 26:854–871
- Charles KK, Luoh MC (2007) Male incarceration, the marriage market and female outcomes. Unpublished Manuscript
- Dahl GB (2010) Early teen marriage and future poverty. *Demography* 47:689–718
- Dougherty C (2005) Why are the returns to schooling higher for women than for men? *J Hum Resour* 40:969–988
- Ellwood DT, Jencks C (2002) The growing difference in family structure. What do we know? Where do we look for answers? Unpublished manuscript
- Field E, Ambrus A (2008) Early marriage, age of menarche, and female schooling attainment in Bangladesh. *J Polit Econ* 116:881–930
- Gladden T (1999) Labor market effects of family ties and migration. Unpublished Manuscript
- Goldin C, Katz LF (2002) The power of the pill: oral contraceptives and women's career and marriage decisions. *J Polit Econ* 110:730–770

- Goldin C, Katz LF, Kuziemko I (2006) The homecoming of american college women: the reversal of the college gender gap. *J Econ Perspect* 20:133–156
- Henderson DJ, Polachek SW, Wang L (2011) Heterogeneity in schooling rates of return. *Econ Educ Rev* 30:1202–1214
- Hersch J, Stratton LS (1997) Housework, fixed effects, and wages of married workers. *J Hum Resour*:32
- Hersch J, Stratton LS (2000) Household specialization and the male marriage wage premium. *Ind Labor Relat Rev* 54:78–94
- Huber M, Mellace G (2014) Testing exclusion restrictions and additive separability in sample selection models. *Empir Econ* 2011–45
- Imbens G, Angrist J (1994) Identification and estimation of local average treatment effects. *Econometrica* 62:467–475
- Keeley M (1974) A model of marital formation: the determinants of the optimal age at first marriage. Ph.D. thesis, University of Chicago
- Kelejian HH (1971) Two-stage least squares and econometric systems linear in parameters but nonlinear in the endogeneous variables. *J Am Stat Assoc* 66:373–74
- Kerckhoff AC, Parrow AA (1979) The effect of early marriage on the educational attainment of young men. *J Marriage Fam* 41:97–107
- Koenker R, Hallock KF (2001) Quantile regression. *J Econ Perspect* 15:143–156
- Korenman S, Neumark D (1992) Marriage, motherhood, and wages. *J Hum Resour* 27:233–55
- Lam D (1988) Marriage markets and assortative mating with household public goods: theoretical results and empirical implications. *J Hum Resour* 23:462–487
- Loughran D (2002) The effect of male wage inequality on female age at first marriage. *Rev Econ Stat* 84:237–250
- Loughran DS, Zissimopoulos J (2004) Are there gains to delaying marriage? The effect of age at first marriage on career development and wages. Unpublished Manuscript
- Loughran DS, Zissimopoulos J (2009) Why wait? The effect of marriage and childbearing on the wage growth of men and women. *J Hum Resour* 44:326–349
- Low H, Meghir C, Pistaferri L (2010) Wage risk and employment risk over the life cycle. *Am Econ Rev* 100:1432–67
- Lundberg S, Pollak RA (1995) Bargaining and distribution in marriage. *J Econ Perspect* 10:139–58
- Lundberg SJ, Rose E (2000) Parenthood and the earnings of married men and women. *Labour Econ* 7:689–710
- Lundberg SJ, Rose E (2002) The effects of sons and daughters on men's labor supply and wages. *Rev Econ Stat* 84:251–268
- Martins M (2001) Parametric and semiparametric estimation of sample selection models: an empirical application to the female labour force in Portugal. *J Appl Econom* 16:23–39
- Miller AR (2011) The effects of motherhood timing on career path. *J Popul Econ* 24:1071–1100
- Miller AR (2013) Lifecycle events and their consequences: job loss, family change, and declines in health, chapter marriage timing, motherhood timing, and women's wellbeing in retirement. Stanford University Press, Stanford, pp 109–132
- Mroz T (1987) The sensitivity of an empirical model of married women's hours of work to economic and statistical assumptions. *Econometrica* 55:765–799
- Polachek SW, Siebert WS (1993) The economics of earnings. Cambridge University Press, Cambridge
- Rose E (2001) Marriage and assortative mating: how have the patterns changed? University of Washington. Working Paper pp. 0–25
- Ruggles S, Alexander JT, Genadek K, Goeken R, Schroeder MB, Sobek M (2010) Integrated public use microdata series: version 5.0 [machine-readable database]. Technical report, University of Minnesota
- Shemyakina O (2007) The effect of armed conflict in Tajikistan on the marriage market and female reproductive behavior. Unpublished Manuscript
- Staiger D, Stock JH (1997) Instrumental variables regression with weak instruments. *Econometrica* 65:557–586
- Wooldridge JM (2002) Econometric analysis of cross section and panel data. MIT Press
- Wooldridge JM (2010) Econometric analysis of cross section and panel data. MIT Press
- Zabel JE (1993) The relationship between hours of work and labor force participation in four models of labor supply behavior. *J Labor Econ* 11:387–416
- Zhang J (1995) Do men with higher wages marry earlier and later? *Econ Lett* 49:193–196